

Summer 2026 Financial Engineering Research Projects

Project Overview

QuantStrat is partnering with the Financial Engineering Department at the University of Illinois to mentor this team-based research replication project. QuantStrat is a global proprietary quantitative trading firm and quantitative asset manager formed by Brian Peterson in 2019. We will be adapting the research replication process utilized in the FIN 554 class at Illinois, applying the scientific method to work on team replications of quantitative finance papers. Research replication is a key skill for any professional working in quantitative finance.

We've discussed with the Financial Engineering Department that students should form teams of 3-5 people. For non-technical students, we suggest teaming up with at least one peer who has a computer science, engineering, or other technical background. For students with purely computational or engineering backgrounds, we encourage seeking a team member with a finance background to provide context on the topic. Complementary backgrounds are highly beneficial, as they allow teams to leverage a diverse set of skills throughout the project.

Mentors

The project sponsor and lead mentor will be Brian Peterson. Brian has over 25 years of experience in quantitative research and strategy development. He is a prominent researcher with over 1,000 citations and the lead author of over 10 packages for using R in finance, including `quantstrat`, `PerformanceAnalytics`, and `PortfolioAnalytics`. Brian is also the Instructor for FIN 554 Algo Trading in the Financial Engineering program at Illinois.

Our second mentor will be Bryan Rodriguez. Bryan works as a quantitative analyst at QuantStrat and serves as a long-time adjunct instructor in the Economics and Data Science departments at Roosevelt University in Chicago.

We will schedule one regular weekly meeting with each team to discuss the project. Both Brian and Bryan will be available via text and email as well.

Data Availability

Project scope is determined both by team interest and by the availability of data. We have high-frequency cryptocurrency data available as well as access to WRDS for datasets such as the CRSP/Compustat daily equity historical data. We also have high-frequency futures data, but only up until approximately 2019, so cryptocurrency and equities projects are strongly preferred.

Potential Replication Papers

The table below presents candidate papers for replication. Your choices are not limited to only these papers; rather, these papers were selected based on the research interests the students communicated with the Financial Engineering department chair. Consider the topics of interest to you and your team, and feel free to propose papers that are not on this list. We will work with your team to ensure that the papers you select are feasible for replication given the data and time constraints of the project as well as the background of you and your teammates.

Authors	Full Title	URL
Afolabi et al. (2017)	Hierarchical Meta-Learning in Time Series Forecasting for Improved Interference-Less Machine Learning	Link
Ammann, Moellenbeck, and Schmid (2011)	Feasible momentum strategies in the US stock market	Link
Babaei, Giudici, and Raffinetti (2022)	Explainable artificial intelligence for crypto asset allocation	Link
Baltas and Kosowski (2012)	Improving time-series momentum strategies: The role of trading signals and volatility estimators	Link
Bhambhwani, Delikouras, and Korniotis (2019)	Do fundamentals drive cryptocurrency prices?	Link
Biais et al. (2023)	Equilibrium bitcoin pricing	Link
Bollerslev, Litvinova, and Tauchen (2006)	Leverage and volatility feedback effects in high-frequency data	Link
Campisi, Muzzioli, and De Baets (2024)	A comparison of machine learning methods for predicting the direction of the US stock market on the basis of volatility indices	Link
Caporale and Plastun (2020)	Momentum effects in the cryptocurrency market after one-day abnormal returns	Link
Chandak et al. (2019)	Learning action representations for reinforcement learning	Link

Authors	Full Title	URL
Chordia, Subrahmanyam, and Tong (2014)	Have capital market anomalies attenuated in the recent era of high liquidity and trading activity?	Link
Dobrynskaya (2021)	Cryptocurrency Momentum and Reversal	Link
Dudler, Gmuer, and Malamud (2014)	Risk Adjusted Time Series Momentum	Link
Funie, Salmon, and Luk (2014)	A hybrid genetic-programming swarm-optimisation approach for examining the nature and stability of high frequency trading strategies	Link
Gu, Kelly, and Xiu (2020)	Empirical Asset Pricing via Machine Learning	Link
Guilbaud and Pham (2015)	Optimal High-Frequency Trading in a Pro Rata Microstructure with Predictive Information	Link
Han, Yang, and Zhou (2013)	A new anomaly: The cross-sectional profitability of technical analysis	Link
Hong and Lee (2017)	A General Approach to Testing Volatility Models in Time Series	Link
Karassavidis, Kateris, and Ioannidis (2025)	Quantitative Evaluation of Volatility-Adaptive Trend-Following Models in Cryptocurrency Markets	Link
Kearns and Nevmyvaka (2013)	Machine learning for market microstructure and high frequency trading	Link
Labadie, Lehalle, et al. (2010)	Optimal algorithmic trading and market microstructure	Link
Lehalle (2013)	Market Microstructure Knowledge Needed for Controlling an Intra-Day Trading Process	Link
Li et al. (2021)	MAX momentum in cryptocurrency markets	Link
Liu and Tsyvinski (2021)	Risks and returns of cryptocurrency	Link
McLean and Pontiff (2016)	Does Academic Research Destroy Stock Return Predictability?	Link
Shen, Jiang, and Zhang (2012)	Stock market forecasting using machine learning algorithms	Link
Tayal (2009)	Regime Switching and Technical Trading with Dynamic Bayesian Networks in High-Frequency Stock Markets	Link
Tran, Pham-Hi, and Bui (2023)	Optimizing Automated Trading Systems with Deep Reinforcement Learning	Link
Zakamulin and Giner (2022)	Optimal Trend Following Rules in Two-State Regime-Switching Models	Link

AI Usage

We encourage teams to use AI where relevant for assisting their research and coding. We use LLMs every day as part of our research and coding processes, and will specifically discuss AI for Literature Review drafts, data parsing, coding assistance in R and Python (we use both languages extensively), testing, and reproducible documents using technology such as Quarto and BibTeX. All university students have access to free GitHub Copilot accounts, so we can discuss these applications in the context of using Copilot along with other tools like AI-aware IDEs like VS Code and Zed. Additionally, we will use Git for a fully reproducible research flow that students will be able to highlight on their CVs after the project is completed.

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